

Review Article

PROSTATE BIOPSY: INDICATIONS AND TECHNIQUE

BRIAN R. MATLAGA, L. ANDREW ESKEW* AND DAVID L. McCULLOUGH†

From the Department of Urology, Wake Forest University School of Medicine, Winston-Salem, North Carolina

ABSTRACT

Purpose: The last decade has seen numerous modifications in the way prostate cancer is diagnosed. We review the current indications for and methods of prostate biopsy.

Materials and Methods: The English language literature was reviewed regarding major indications for and methods of prostate biopsy. Pertinent peer reviewed articles were collated and analyzed.

Results: The most widely accepted indication for prostate biopsy is a prostate specific antigen (PSA) value of greater than 4.0 ng./ml. However, some investigators advocate prostate biopsy for men with a PSA value in the 2.5 to 4.0 ng./ml. range, believing that use of this parameter results in detection of a greater number of cases of curable disease. Age specific PSA range, percent free PSA and presence of prostatic intraepithelial neoplasia or atypia are all considered to be relative indications for prostate biopsy. The current literature describes a trend toward increasing the number of cores obtained and the sites biopsied beyond those of the standard sextant technique. The additional cores in many series are obtained from more lateral regions of the gland.

Conclusions: Although several criteria are used as indications for initial prostate biopsy, all are based on PSA level and/or abnormal digital rectal examination. Future improvements in currently used prostate cancer markers may result in better selection of cases to biopsy. There is no universally accepted technique of prostate gland biopsy. The current literature supports use of more extensive biopsy techniques to increase the likelihood of prostate cancer detection.

KEY WORDS: biopsy, needle; prostate-specific antigen; prostatic neoplasms; prostate

Diagnosis of prostate cancer requires obtaining cancerous tissue from the prostate gland during biopsy. The advent of transrectal ultrasound has revolutionized prostate biopsy techniques and has greatly increased the diagnostic accuracy of biopsy. The addition of prostate specific antigen (PSA) as a screening tool for prostate cancer has increased the application of transrectal ultrasound and prostate biopsy. With more than 175,000 new cases of prostate cancer diagnosed each year, and many more cases biopsied to rule out carcinoma, transrectal ultrasound with prostate biopsy has become a commonly performed urological procedure. We review the current indications for and methods of prostate biopsy, which have been refined in the last 10 years.

GENERAL INDICATIONS FOR PROSTATE BIOPSY

PSA greater than 4.0 ng./ml. In early studies of prostate cancer detection biopsies were performed for abnormalities on digital rectal examination or transrectal ultrasound but not for PSA abnormalities alone.¹ It was not until 1991 that PSA alone was considered to be an indication for biopsy.² Following the introduction of PSA as an indication for prostate biopsy, there was debate regarding whether patients should undergo biopsy if PSA levels were 4 to 10 ng./ml. and no other risk factors, such as an abnormal digital rectal examination or transrectal ultrasound, were present. In 1992 the cancer detection rate for patients with PSA levels of 4 to 10 ng./ml. and normal digital rectal examination was re-

ported to be 5.5%.³ Recent data suggest that the current cancer detection rate is 20% to 30% for patients with a PSA of 4 to 10 ng./ml.^{2,4} For this reason, serum PSA greater than 4.0 ng./ml. is considered an indication for biopsy.

Abnormal digital rectal examination. Abnormal digital rectal examination is an absolute indication for prostate biopsy. Based on the report of Richie et al, in which 18% of patients with prostate cancer had the disease detected solely by abnormal digital rectal examination, a firm nodule or a diffusely firm prostate should prompt biopsy regardless of serum PSA level.⁵ In 1998 Schroder et al reported that digital rectal examination has a poor predictive value in the detection of prostate cancer and should be replaced with a more sensitive test.⁶ Since that time, the European Randomized Study of Screening for Prostate Cancer has abandoned use of digital rectal examination as a screening study.⁷ However, Carvalhal et al suggest that digital rectal examination be performed in men with a PSA of 1.0 ng./ml. or greater, as an appreciable proportion (14% to 30%) of patients with a suspicious digital rectal examination and a PSA of 1 to 4 ng./ml. will have prostate cancer detected on initial prostate biopsy.⁸ At our institution digital rectal examination is routinely performed, since some patients with a suspicious examination and a low PSA will have aggressive prostate cancer.

Presence of high grade prostatic intraepithelial neoplasia or atypia on prostate biopsy. While low grade prostatic intraepithelial neoplasia is not considered a precancerous lesion, it has been demonstrated that high grade prostatic intraepithelial neoplasia has a high predictive value for carcinoma on subsequent biopsy. Current data suggest that 27% to 79% of patients found to have prostatic intraepithelial neoplasia on

* Financial interest and/or other relationship with Microvasive and Alza Pharmaceuticals.

† Financial interest and/or other relationship with Bayer and Pharmacia.

initial biopsy will have development of adenocarcinoma of the prostate.⁹⁻¹⁵ In addition, a finding of atypia on prostate biopsy confers a high risk of cancer on subsequent biopsy. Chan and Epstein recently reported that up to 49% of patients with a diagnosis of atypia on initial biopsy are subsequently found to have prostate cancer.¹⁶ Park et al similarly reported a 45% positive repeat biopsy rate in men previously diagnosed with atypia.¹⁷ While the optimal time between biopsy sessions in patients with high grade prostatic intraepithelial neoplasia or atypia is unknown, it has been suggested that these patients undergo repeat biopsies at 3 to 12-month intervals with extensive biopsy technique.¹⁸

RELATIVE INDICATIONS FOR PROSTATE BIOPSY

Palpable abnormality or increasing PSA after definitive therapy. A palpable abnormality and/or increasing PSA after curative therapy for cancer (radical prostatectomy, radiation therapy, cryotherapy and so forth) is a relative indication for prostate biopsy to rule out local recurrence. For patients who have undergone radical prostatectomy this finding may include an anastomotic biopsy.¹⁹ For patients who have undergone radiation or cryotherapy a formal prostate biopsy is indicated if absolute proof of recurrent or persistent cancer is required.²⁰ The usefulness of a biopsy in this setting is not clear, as some clinicians believe that an increasing PSA following these therapies provides sufficient evidence of persistent cancer. Recent evidence presented by Koppie et al suggests that decisions regarding local radiation therapy for increasing PSA following radical prostatectomy may be safely and effectively made without documenting recurrent local disease.²¹

Age adjusted PSA increase. Age adjusted PSA determinations were introduced as a means to improve cancer detection sensitivity in younger men and specificity in older men. One study revealed a significant increase in the number of organ confined cancers detected in men younger than 50 years (all of whom had a normal digital rectal examination) when age specific PSA reference ranges were used (table 1).⁴ However, Catalona et al found that increasing the PSA cutoff to 4.5 ng./ml. among men 60 to 69 years old would result in 15% fewer biopsy sessions but would miss 8% of organ confined cancers.²² Increasing the PSA cutoff to 6.5 ng./ml. among men 70 years or older would result in 44% fewer biopsy sessions but would miss 47% of organ confined cancers.

Low percent free PSA. Serum PSA exists in multiple forms, although most is complexed to protease inhibitors such as α 1-antichymotrypsin.²³ Percent free PSA is lower in patients with than in those without prostate cancer.²⁴ Therefore, percent free PSA may aid in determining which patients should undergo prostate biopsy and especially which patients should undergo repeat biopsy. Catalona et al showed that percent free PSA can improve the specificity of PSA testing for men with a total serum PSA of 2.5 to 10.0 ng./ml.²⁵ They reported that to maintain 95% sensitivity a cutoff of 20% was needed for men 50 to 59 years old, 26% for men 60 to 69 years old and 28% for men 70 to 75 years old. Ultimately a free PSA cutoff of 25% has been recommended to maximize cancer detection in younger men while avoiding unnecessary biopsies in older men.^{25, 26}

PSA velocity. One possible method to make PSA measurement more efficient as a screening study is to use serial measurements to distinguish benign from malignant disease. PSA velocity measures change with time, allowing for the longitudinal assessment of PSA levels. The assumption of this practice is that men with clinically significant prostate cancer will have relatively rapid increases in serum PSA compared to those without prostate cancer. In 1992 Carter et al introduced the concept of PSA velocity and determined that a velocity of greater than 0.75 ng./ml. per year was suggestive of cancer.²⁷ However, a subsequent prospective study by Smith and Catalona found that a cutoff of 0.75 ng./ml. per year for men with a serum PSA of 4.0 ng./ml. or less yielded a sensitivity of 79% and a specificity of 66%.²⁸ For men with a serum PSA of greater than 4.0 ng./ml. a sensitivity of 63% and a specificity of 62% were reported. These data suggest that the optimal use of PSA velocity is in a patient population with a PSA of 4.0 ng./ml. or less.

PSA density. PSA density refers to serum PSA divided by prostate volume in cubic centimeters, which yields the amount of PSA present relative to the amount of prostate tissue present. PSA density was developed in an effort to control for prostate size, recognizing that benign prostatic enlargement, without the presence of cancer, can result in PSA increases.²⁹ PSA density is thought to improve the specificity of PSA testing, as an increased PSA density indicates an increased likelihood of prostate cancer being present. Seaman et al determined that a normal PSA density for men with PSA levels of 4.0 to 10.0 ng./ml. is less than 0.15.³⁰

There are several disadvantages to using PSA density in a clinical setting. An accurate prostate volume is required, which necessitates an invasive transrectal ultrasound. In addition, transrectal ultrasound is often performed during a prostate biopsy session after the decision has already been made to biopsy the gland, which limits the clinical usefulness of PSA density in terms of pre-biopsy medical decision making. There is also a considerable overlap of patients having a PSA density of less than 0.15 who may have benign prostatic hyperplasia, prostatic intraepithelial neoplasia, prostatitis or cancer.³¹ Brawer et al compared serum PSA alone to PSA density in 218 men undergoing prostate biopsy and found that PSA density did not enhance the ability of serum PSA to predict the presence of carcinoma.³² Catalona et al also reported that percent free PSA yielded results similar to those of PSA density without the need for transrectal ultrasound.³³

RECENTLY SUGGESTED INDICATIONS FOR PROSTATE BIOPSY

PSA 2.5 to 4.0 ng./ml. (lower "normal" PSA). Because a significant proportion of men with PSA in the 2.5 to 4.0 ng./ml. range will have prostate cancer, some investigators have advocated decreasing PSA cutoffs to enhance prostate cancer detection.³⁴ Catalona et al detected cancer in 73 of 332 men (22%) with PSA levels of 2.6 to 4.0 ng./ml.³⁵ Smith et al similarly reported an overall cancer detection rate of 27% in men with PSA levels of 2.6 to 4.0 ng./ml.³⁶ It has also been reported that prostate cancer may be detected within 3 to 5 years in 13% to 20% of men with an initial PSA of 2.5 to 4.0 ng./ml., and approximately 80% of patients with cancer and PSA levels of 2.5 to 4.0 ng./ml. are described as having

TABLE 1. Age specific reference ranges for serum PSA concentration, prostatic volume and PSA density

Parameter	Age Group (yrs.)			
	40-49	50-59	60-69	70-79
Serum PSA concentration (ng./ml.)	0.0-2.5	0.0-3.5	0.0-4.5	0.0-6.5
Prostatic volume (ml.)	13-51	15-60	17-70	20-82
PSA density (ng./ml.)	0.0-0.08	0.0-0.10	0.0-0.11	0.0-0.13

For serum PSA concentration and PSA density upper limit of normal was defined as 95th percentile for midpoint of age range from regression analysis. For prostatic volume 97.5th percentile was used to define upper limit. Lower limit was set at 0.0 for PSA concentration and PSA density, and 2.5th percentile for prostatic volume. Reprinted with permission.⁴

clinically significant disease with aggressive characteristics.^{7,37,38} In contrast to these findings, Carter et al found a minimal difference (2% to 4%) in the probability of detecting curable cancer at PSA levels of 2.5 to 6.0 ng/ml. for all age groups.^{39,40} Thus, relative indications for prostate biopsy in the patient with a PSA of 2.5 to 4.0 ng/ml. may include family history of prostate cancer, increased age adjusted PSA and abnormal physical examination.

TECHNIQUE

Patient preparation. The ideal pre-biopsy preparation is efficacious, economical and simple. Although no randomized study has demonstrated the value of a cleansing enema before prostate biopsy, most urologists routinely include this procedure.⁴¹ In addition, pre-biopsy antibiotic prophylaxis has been reported to decrease the rate of infectious complications.^{1,42,43} However, there is currently wide discordance among practicing urologists regarding pre-biopsy preparation. A review by Shandera et al of 568 United States urologists revealed the use of 11 different antibiotics with 23 different timing/duration regimens, resulting in 253 possible combinations.⁴⁴ This number increases when one considers whether an enema was performed and what antibiotic dosage was used. We routinely use a 3-day course of a quinolone antibiotic beginning the day before biopsy. Patients with valvular heart disease are administered parenteral antibiotics as outlined by the American Heart Association. Cleansing enemas are administered the night before and the morning of the procedure.

In the past many urologists thought that the discomfort associated with prostate biopsy was so mild that local anesthesia was not necessary. However, 65% to 90% of patients undergoing transrectal ultrasound guided biopsy of the prostate claim to experience discomfort.^{45,46} In response to these reports there has been interest in finding an effective local anesthetic to allow for less painful biopsy. Nash et al⁴⁷ described transrectal ultrasound guided prostatic nerve blockade with 1% lidocaine, while Issa et al⁴⁸ described a local anesthetic technique using 2% lidocaine jelly instilled intrarectally before transrectal ultrasound guided prostate biopsy. A recent randomized trial has shown that periprostatic nerve block with 1% lidocaine provides anesthesia superior to that achieved with intrarectal lidocaine jelly.⁴⁹ Finally, intravenous sedation can be used in patients requiring more extensive biopsy sessions.^{50,51}

Sextant technique. Introduced by Hodge et al in 1989, the sextant technique has been the worldwide gold standard prostate biopsy method.⁵² As originally described, the sextant method takes 6 biopsies of the prostate, with specimens obtained from the base, middle and apex of the prostate in the mid lobar parasagittal planes bilaterally.

Although the sextant method was the first described systematic method of prostate biopsy, other methods have demonstrated that the sextant technique as originally described has a false-negative rate of approximately 30%.⁵³ This false-negative rate is understandable as this method not only involves taking fewer cores, but also samples a smaller percentage of the peripheral zone of the prostate, where 80% of prostatic adenocarcinomas are found. More recently the sextant method has been modified to include more laterally directed biopsies.⁵⁴ These biopsies, which have been described as "lateral biopsies" in several studies, include areas of the prostate that have been referred to as the "far lateral region" and "anterior horn."

Increasing the number of core biopsies. Multiple in vivo studies have revealed that increasing the number of prostate biopsies enhances prostate cancer detection,^{50,51,53,55,56} while 1 study failed to show this phenomenon.⁵⁷ Importantly Eskew⁵⁸ and Chan⁵⁹ et al recently found that increased sampling of the prostate through extended biopsy techniques

does not increase the detection of potentially insignificant tumors, and it appears to detect earlier stage cancer.

Five region prostate biopsy. Because the cross-sectional anatomy of the prostate demonstrates an almost 3-fold increase in the peripheral zone of the lateral prostate compared with other areas of the gland, a more extensive method of biopsy was developed to sample the lateral area more thoroughly. Eskew et al performed the first prospective study comparing the sextant biopsy method to an alternative form of biopsy.⁵⁰ The 5 region method involves obtaining cores from the traditional sextant regions (6 cores) as well as 4 cores (2 from each side) from the far lateral regions ("anterior horn" or "lateral" biopsies) and 3 midline biopsies.^{55,56} Thus, a total of 13 biopsies are obtained (fig. 1). When gland size is greater than 50 cc, an extra core from each region is often obtained. By increasing the number of peripheral zone biopsies, the 5 region technique decreases the false-negative rate by 35%. Eighty-eight percent of the additionally detected cancers are found in the far lateral regions of the prostate gland, which are composed entirely of peripheral zone tissue. More recently this method of biopsy has included taking 3 cores from all 5 biopsy regions. Other studies have used modifications of this technique.

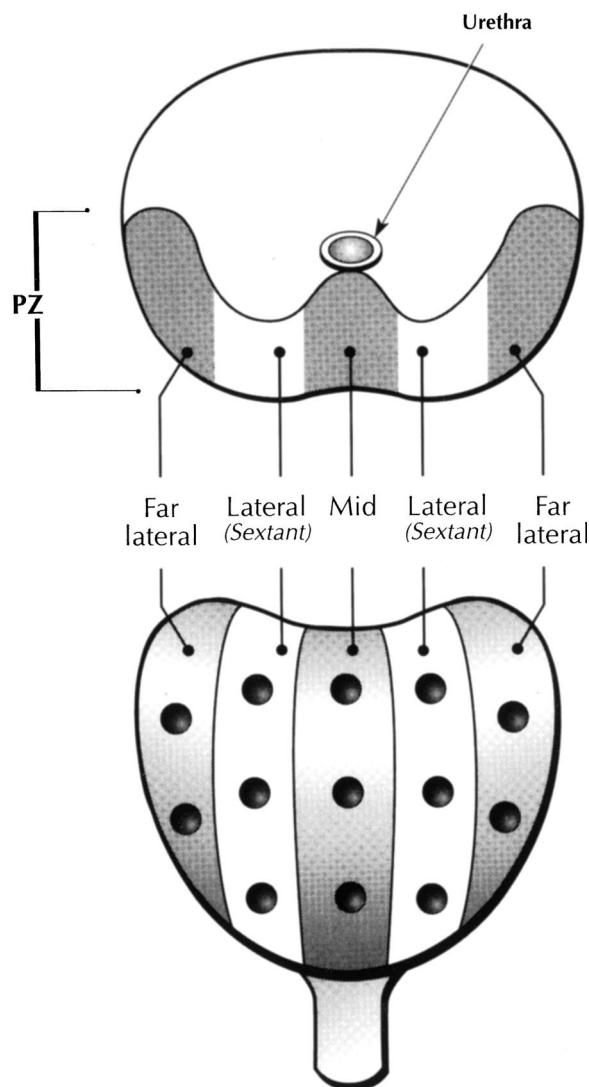


FIG. 1. Cross-sectional and posterior views of prostate reveal peripheral zone (PZ) and areas of biopsy (circles). Shaded areas represent additional regions of biopsy using 5 region method. Reprinted with permission.⁵⁸

Two consecutive sextant biopsies. Levine et al investigated performing 2 consecutive sextant biopsies in a single biopsy setting.⁵³ A total of 137 patients were enrolled in this study. The second set of biopsies provided clinically relevant information in 28% of patients as related to detection of high grade prostatic intraepithelial neoplasia or prostate cancer. In addition, the second set of biopsies increased the number of cancers detected by 30%.

Eleven-core multisite directed biopsy. Babaian et al investigated an 11-core biopsy method in 362 patients, which included sextant biopsies as well as 1 biopsy from the far lateral region (anterior horn), midline and bilateral transition zones (adjacent and anterior to the urethra) (fig. 2).⁵⁵ This study revealed an overall increase of 33% in the prostate cancer detection rate, which was statistically significant. The authors concluded that an extended method of biopsy was tolerated without conscious sedation, and enhanced cancer detection in men undergoing repeat biopsy. This method did not increase cancer detection significantly in patients who were undergoing an initial biopsy. However, only 1 core from the far lateral region and midline was obtained, which may account for this phenomenon.

Eight systematic biopsies. Presti et al evaluated systematic prostate biopsy in 483 patients.⁵⁶ This biopsy scheme obtains 10 samples from the prostate, 6 samples from the traditional sextant regions and 2 cores from each of the lateral regions (peripheral zones). This method is similar to the 5 region technique, with the exception that the midline biopsies are eliminated. The authors found that systematic cores taken from the traditional sextant region, and lateral base and lateral mid portion of the gland could detect 96% of the cancers diagnosed. Eliminating the mid lobar base core of the sextant biopsy, resulting in the 8-biopsy peripheral zone regimen, decreased cancer detection by only 1% (fig. 3). It is unknown whether 8 systematic biopsies is the optimal number. However, this study is useful in determining which cores can be safely omitted (sextant mid lobar base) without significantly affecting the cancer detection rate.⁵⁶ In this study the traditional sextant biopsies missed 20% of the cancers diagnosed. As other studies have shown that cancers are also found at midline aspects of the gland, this biopsy technique may have a higher false-negative rate than other techniques reported.^{50, 55}

Saturation biopsy technique for repeat biopsy. Stewart et al used a saturation prostate biopsy technique to evaluate 224 patients who previously had negative sextant biopsy results and persistent indications for repeat biopsy.⁵¹ The repeat saturation biopsies were performed with the patients under general anesthesia, regional anesthesia or intravenous sedation. An average of 23 cores (range 14 to 45) were obtained during a biopsy session. Using an inward radial step ap-

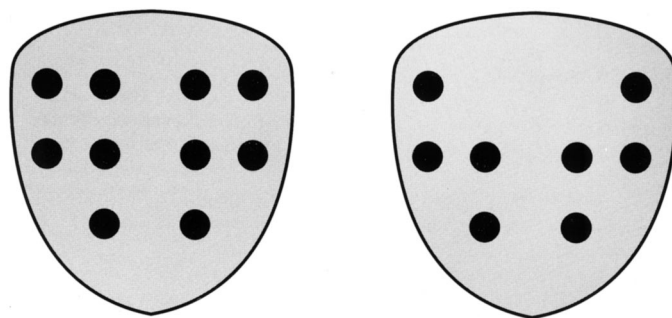


FIG. 3. Initial 10-core technique and proposed final octet technique as recommended by Presti et al.⁵⁶ Reprinted with permission [Cookson, M.S.: Technical refinements of transrectal ultrasound-guided needle biopsy and repeat biopsy of the prostate. AUA Update Series, vol. 20, lesson 11, pp. 82–88, 2001].

proach, prostates were biopsied starting at the far lateral peripheral zone (anterior horn) and continuing until the mid gland was reached. The process was then repeated on the contralateral side. Larger prostates required more samples. Interestingly 34% of patients were found to have cancer with this technique. This result compares to 38.4% of patients who had previously undergone negative sextant biopsy who were found to have prostate cancer on repeat biopsy using the 5 region technique.⁶⁰ Median time from first negative sextant biopsy to positive saturation biopsy was 2.4 years. The location where cancer was ultimately detected within the prostate gland would be interesting to know but was not reported in this study.

Relationship of prostate biopsy and gland volume in cancer detection. There is an inverse relationship between prostate size and the likelihood of finding prostate cancer in a sampling.^{61–63} The relative amount of gland that is sampled relies directly on the size of the gland and, thus, the ideal number of cores to take may be dependent on the size of the gland as calculated by transrectal ultrasound. Gland size can affect the yield of prostate biopsy, producing a higher false-negative rate in larger glands.

Stricker et al used the Bayesian theory to demonstrate that for a fixed percentage volume of prostate cancer the probability of finding cancer increases as the number of biopsies increases.⁶⁴ This mathematically derived hypothesis supposes that all prostates are of the same volume and, therefore, there is always a fixed percentage volume of cancer for any prostate volume. This approach indicates a bias for finding cancer in smaller glands while the same volume of cancer in a larger gland (smaller percentage volume of cancer) might otherwise remain undetected. If the volume of cancer in a gland is used for clinical decision making, more core samples should be taken when sampling larger glands.

The impact of prostate volume on prostate biopsy results is well described. Uzzo et al reported achieving a cancer detection rate of 23% in glands 50 cc or greater versus 38% in glands less than 50 cc using the sextant biopsy technique.⁶¹ Levine et al also evaluated the positive rate of sextant biopsy according to gland size.⁵³ In their study population cancer was detected in 43%, 27% and 24% of men with prostate volumes less than 30 cc, 30 to 50 cc and greater than 50 cc, respectively. Applewhite et al used the 5 region technique and reported prostate cancer detection in 50%, 41% and 31% of men with prostate volumes less than 30 cc, 30 to 50 cc and greater than 50 cc, respectively.⁶⁵

Repeat biopsy population. The decision regarding whether to perform a repeat biopsy in a patient with persistent indications is a common problem faced by urologists. Low yields from repeat sextant biopsies have been reported in many series, and the cancers missed in an initial biopsy are not necessarily small indolent tumors.^{66–69} These cases present

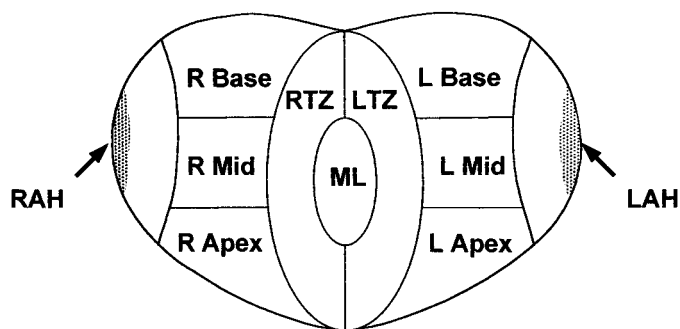


FIG. 2. Schema of prostate depicts 11 sites of biopsy strategy. *R Base*, right base (sextant). *R Mid*, right mid (sextant). *R Apex*, right apex (sextant). *L Base*, left base (sextant). *L Mid*, left mid (sextant). *L Apex*, left apex (sextant). *RAH*, right anterior horn. *LAH*, left anterior horn. *RTZ*, right transition zone. *LTZ*, left transition zone. *ML*, midline. Reprinted with permission.⁵⁵

TABLE 2. Results of repeat biopsies

References	No. Pts.	Mean No. Biopsies	% Ca	Mean Interval Between Biopsies (mos.)
Keetch et al ⁶⁸	427	4-6	19	12
Ellis and Brawer ⁶⁷	100	6	20	15
Babaian et al ⁵⁵	277	11	29	16
Borboroglu et al ⁷⁰	57	22.5	30	10
Stewart et al ⁵¹	224	23	34	28.8
Applewhite et al ⁶⁰	125	17.6	38.4	24.6

a unique challenge to the urologist. Importantly, for the patient with a prior diagnosis of prostatic intraepithelial neoplasia or atypia Park et al reported that prostate cancer was detected not only at sites adjacent to the original abnormal finding, but also at remote sites.¹⁷

Applewhite et al used the 5 region biopsy technique in a repeat biopsy population.⁶⁰ The positive biopsy rate was 38.4% for patients who had previously undergone sextant biopsy, which compares favorably to the 34% cancer detection rate described by Stewart et al using a saturation technique.⁵¹ The positive biopsy rate for those who had previously undergone a 5 region biopsy was 30%. For patients who had previously undergone sextant biopsy, the 5 region technique had a relative increase in yield of 33% over the sextant technique. For those who had previously undergone biopsy with the 5 region technique a repeat biopsy with the same technique yielded a relative increase of 39% over the sextant technique. The overall positive biopsy rate for patients who had previously undergone biopsy by either the sextant or the 5 region technique was 34.4%, and mean number of biopsy cores obtained was 17.6 (table 2).^{51, 55, 60, 67, 68, 70}

Babaian et al reported the results of a systematic 11-core technique in a repeat biopsy population of 277 men.⁵⁵ The technique consisted of standard sextant sampling with the addition of far lateral, midline and transition zone biopsies. The overall cancer detection rate was 29% for this population. Thirty-three percent of cancers were found in the additional (nonsextant) regions only.

Borboroglu et al recently reported their experience using an extensive prostate biopsy technique, averaging 22.5 cores in patients after previously benign sextant biopsies.⁷⁰ Biopsies were obtained from each of 6 sagittal regions, including the far lateral and mid transition zones. Overall yield was 30%. Additional details of repeat biopsy studies are reported in table 2. Interestingly, although the optimal number of biopsy cores is unknown, the greatest yield appears to be offered by 11 to 18 biopsies.

Perrotti et al used endorectal magnetic resonance imaging (MRI) in an effort to improve prostate cancer detection rate on repeat biopsy.⁷¹ Following endorectal MRI, regions of suspicion were mapped on a 3-dimensional prostate model. Subsequent transrectal ultrasound guided prostate biopsy was performed, with ultrasound imaging correlated with the images from MRI. Overall endorectal MRI had a 40% positive predictive value and 70% accuracy, making this technique a possible method to improve prostate cancer detection.

The role of transition zone biopsies. Although approximately 20% of prostate cancers originate in the transition zone, isolated transition zone tumors detected on prostate biopsy are uncommon. The addition of transition zone biopsies to the initial biopsy strategy increases detection rates by only 1.8% to 4.3%, and there are few data to support the recommendation for routine transition zone sampling.⁷²⁻⁷⁵ However, in men undergoing repeat biopsy the yield of malignancy from the transition zone is higher, ranging from 10% to 13%.^{76, 77} Thus, transition zone biopsies may be indicated in patients in whom systematic sextant biopsies fail to reveal cancer but whose PSA is markedly increased or rapidly increasing.

In men with previously negative biopsy results the important question is whether the undiagnosed cancer is in the transition zone. Keetch and Catalona found a yield of 10% from transition zone biopsies in patients undergoing repeat biopsy.⁷⁶ However, certain subsets of patients may have a much higher incidence of transition zone carcinoma. In repeat biopsies in men with a mean PSA of 32 ng/ml., normal digital rectal examination and a clinical picture suspicious for carcinoma Lui et al found that 53% of cancers were detected in the transition zone only.⁷⁷ Therefore, it is reasonable to include transition zone biopsies in the repeat biopsy population.

Chen et al found that the highest detection rate for transition zone biopsies in a computer generated model was observed when the biopsies were initiated near the prostatic apex and the needles were inserted to a depth of 3 cm. before firing.⁷⁸ Lower rates of detection were noted when the needles were inserted to a depth of 1 to 2 cm. and as one moved more toward the base of the gland.

Prostate biopsy in patients without a rectum. Biopsy in patients who have lost the rectum to malignancy or inflammatory bowel disease presents a diagnostic challenge in terms of screening for prostate cancer. The indication for prostate biopsy is increased PSA in this population, as digital rectal examination is not possible.

Two techniques have been described for patients without a rectum, namely transurethral ultrasound guided biopsy and transperineal ultrasound guided biopsy. Seaman et al used transurethral ultrasound guidance to perform transperineal prostate biopsy in 5 men who had lost the rectum due to total colectomy or abdominoperineal resection.⁷⁹ Seven biopsy procedures were performed in this population. Cancer was diagnosed in 3 of the 5 patients using this technique. Sextant biopsies as well as ultrasound directed lesional biopsies were performed. Good visualization of the prostate was reported. The authors concluded that transurethral ultrasound enabled accurate sonographic assessment and precise prostate biopsy in patients without access to the rectum.

Shinghal and Terris evaluated transperineal ultrasound guided prostate biopsy in 20 men who had previously been diagnosed with prostate cancer by transrectal ultrasound guided biopsy.⁸⁰ Prostate ultrasound and subsequent biopsies were performed transperineally, mimicking the clinical scenario of performing a biopsy in a patient without a rectum. Using this technique, only 2 of the 20 patients were diagnosed with cancer, yielding a sensitivity of only 10%. The authors concluded that transperineal guided prostate biopsies are less accurate than transrectal ultrasound guided biopsies, and that this finding emphasizes the need for prostate cancer screening before removal of the rectum.

Complications. The transrectal approach to prostate biopsy has been associated with complications ranging from dysuria to death. Fortunately major complications, categorized as those requiring hospitalization, are uncommon. Rodríguez and Terris reported only 1 major complication in 77 patients (1.3%) undergoing prostate biopsies, which involved a vasovagal episode with seizures at the time of biopsy.⁴² This rate is consistent with other published series.^{45, 81} However, the authors also reported 135 minor complications, including discomfort, vasovagal episode, rectal bleeding and hematuria.⁴²

Hemorrhagic complications are the most commonly reported sequelae of prostate biopsy, with gross hematuria being the most frequent, occurring in up to 50% of patients. Obtaining biopsies in the midline of the prostate may increase the incidence of this complication. During the course of the procedure, hemorrhage may be visualized within the bladder following needle biopsy. If this is the case, or if gross blood is noted at the urethral meatus, bladder catheterization and clot irrigation may be indicated. At the conclusion of the procedure digital rectal examination can be performed to

confirm the absence of significant rectal hemorrhage. If significant rectal bleeding is encountered, a well lubricated vaginal pack may be placed in the rectal space for several hours to provide tamponade of the bleeding site. Alternatively endoscopy of the rectum for the treatment of severe rectal bleeding following prostate biopsy has been described.⁸²

Fortunately infectious complications are rare, accounting for just 2.5% of the complications in the series of Rodríguez and Terris.⁴² Antibiotic prophylaxis ostensibly decreases the likelihood of infection. Furthermore, individuals with infectious complications, such as fever, chills and urinary tract infection, can often be treated as outpatients.

CONCLUSIONS

Diagnosis of prostate cancer requires obtaining cancerous tissue from the prostate gland. Transrectal ultrasound guided biopsy has emerged as the preferred method and is widely used in the diagnosis of prostate cancer. In an appropriately screened population general indications for prostate biopsy include abnormal digital rectal examination, PSA greater than 4.0 ng/mL, presence of high grade prostatic intraepithelial neoplasia or atypia on prior biopsy, and palpable abnormality and/or increasing PSA after "curative" therapy for prostate cancer. Relative indications for biopsy include age adjusted PSA increase, low percent free PSA (25% or less) and a PSA velocity of greater than 0.75 ng/mL per year. Decreasing the PSA biopsy cut point to 2.5 ng/mL to increase the likelihood of detecting organ confined disease has been advocated by some investigators. However, this lower cut point has not yet been widely accepted as an indication for prostate biopsy. In addition, standard patient preparation for biopsy varies. However, pre-biopsy enemas and antibiotics are commonly used.

Once the gold standard method of biopsy, the sextant technique has been shown by several studies to yield a false-negative rate of approximately 30%. Contemporary biopsy strategies have consistently decreased the false-negative rate of the sextant biopsy by taking additional laterally directed samples. Larger prostate glands may require additional biopsies to detect clinically significant cancer.

The literature supports increasing the number of cores in the repeat biopsy population to maximize prostate cancer diagnosis. It is rare to find isolated disease in transition zone biopsies, although biopsy of this area may in some cases demonstrate cancer. Finally, there is a need to standardize the method and nomenclature of extended biopsy technique in terms of reading and understanding the literature involving prostate biopsy.

REFERENCES

- Cooner, W. H., Mosley, B. R., Rutherford, C. L., Jr., Beard, J. H., Pond, H. S., Terry, W. J. et al: Prostate cancer detection in a clinical urological practice by ultrasonography. Digital rectal examination and prostate specific antigen. *J Urol*, **143**: 1146, 1990
- Catalona, W. J., Smith, D. S., Ratliff, T. L., Dodds, K. M., Coplen, D. E., Yuan, J. J. et al: Measurement of prostate-specific antigen in serum as a screening test for prostate cancer. *N Engl J Med*, **324**: 1156, 1991
- Stamey, T. A. and McNeal, J. E.: Adenocarcinoma of the prostate. In: Campbell's Urology, 6th ed. Edited by P. C. Walsh, A. B. Retik, T. A. Stamey and E. D. Vaughan, Jr. Philadelphia: W. B. Saunders Co., 1992
- Oesterling, J. E., Jacobsen, S. J., Chute, C. G., Guess, H. A., Girman, C. J., Panser, L. A. et al: Serum prostate-specific antigen in a community-based population of healthy men. Establishment of age-specific reference ranges. *JAMA*, **270**: 860, 1993
- Richie, J. P., Catalona, W. J., Ahmann, F. R., Hudson, M. A., Scardino, P. T., Flanigan, R. C. et al: Effect of patient age on early detection of prostate cancer with serum prostate-specific antigen and digital rectal examination. *Urology*, **42**: 365, 1993
- Schroder, F. H., van der Maas, P., Beemsterboer, P., Kruger, A. B., Hoedemaeker, R., Rietbergen, J. et al: Evaluation of the digital rectal examination as a screening test for prostate cancer. Rotterdam section of the European Randomized Study of Screening for Prostate Cancer. *J Natl Cancer Inst*, **90**: 1817, 1998
- Schröder, F. H., van der Cruyssen-Koeter, I., de Koning, H. J., Vis, A. N., Hoedemaeker, R. F. and Kranse, R.: Prostate cancer detection at low prostate specific antigen. *J Urol*, **163**: 806, 2000
- Carvalhal, G. F., Smith, D. S., Mager, D. E., Ramos, C. and Catalona, W. J.: Digital rectal examination for detecting prostate cancer at prostate specific antigen levels of 4 ng/mL or less. *J Urol*, **161**: 835, 1999
- Abouseif, S., Shinohara, K., Weidner, N., Narayan, P. and Carroll, P. R.: The significance of prostatic intra-epithelial neoplasia. *Br J Urol*, **76**: 355, 1995
- Raviv, G., Janssen, T., Zlotta, A. R., Descamps, F., Verheest, A. and Schulman, C. C.: Prostatic intraepithelial neoplasia: influence of clinical and pathological data on the detection of prostate cancer. *J Urol*, **156**: 1050, 1996
- Langer, J. E., Rovner, E. S., Coleman, B. G., Yin, D., Arger, P. H., Malkowicz, S. B. et al: Strategy for repeat biopsy of patients with prostatic intraepithelial neoplasia detected by prostate needle biopsy. *J Urol*, **155**: 228, 1996
- Wills, M. L., Hamper, U. M., Partin, A. W. and Epstein, J. I.: Incidence of high-grade prostatic intraepithelial neoplasia in sextant needle biopsy specimens. *Urology*, **49**: 367, 1997
- O'dowd, G. J., Miller, M. C., Orozco, R. and Veltri, R. W.: Analysis of repeated biopsy results within 1 year after a non-cancer diagnosis. *Urology*, **55**: 553, 2000
- Davidson, D., Bostwick, D. G., Qian, J., Wollan, P. C., Oesterling, J. E., Rudders, R. A. et al: Prostatic intraepithelial neoplasia is a risk factor for adenocarcinoma: predictive accuracy in needle biopsies. *J Urol*, **154**: 1295, 1995
- Berner, A., Danielsen, H. E., Pettersen, E. O., Fossa, S. D., Reith, A. and Nesland, J. M.: DNA distribution in the prostate. Normal gland, benign and premalignant lesions, and subsequent adenocarcinomas. *Anal Quant Cytol Histol*, **15**: 247, 1993
- Chan, T. Y. and Epstein, J. I.: Follow-up of atypical prostate needle biopsies suspicious for cancer. *Urology*, **53**: 351, 1999
- Park, S., Shinohara, K., Grossfeld, G. D. and Carroll, P. R.: Prostate cancer detection in men with prior high grade prostatic intraepithelial neoplasia or atypical prostate biopsy. *J Urol*, **165**: 1409, 2001
- Iczkowski, K. A. and Bostwick, D. G.: Prostate biopsy interpretation. Current concepts, 1999. *Urol Clin North Am*, **26**: 435, 1999
- Trapasso, J. G., deKernion, J. B., Smith, R. B. and Dorey, F.: The incidence and significance of detectable levels of serum prostate specific antigen after radical prostatectomy. *J Urol*, **152**: 1821, 1994
- Foster, L. S., Jajodia, P., Fournier, G., Jr., Shinohara, K., Carroll, P. and Narayan, P.: The value of prostate specific antigen and transrectal ultrasound guided biopsy in detecting prostatic fossa recurrences following radical prostatectomy. *J Urol*, **149**: 1024, 1993
- Koppie, T. M., Grossfeld, G. D., Nudell, D. M., Weinberg, V. K. and Carroll, P. R.: Is anastomotic biopsy necessary before radiotherapy after radical prostatectomy? *J Urol*, **166**: 111, 2001
- Catalona, W. J., Hudson, M. A., Scardino, P. T., Richie, J. P., Ahmann, F. R., Flanigan, R. C. et al: Selection of optimal prostate specific antigen cutoffs for early detection of prostate cancer: receiver operating characteristic curves. *J Urol*, **152**: 2037, 1994
- Lilja, H., Christensson, A., Dahlen, U., Matikainen, M. T., Nilsson, O., Pettersson, K. et al: Prostate-specific antigen in serum occurs predominantly in complex with alpha 1-antichymotrypsin. *Clin Chem*, **37**: 1618, 1991
- Catalona, W. J., Smith, D. S., Wolfert, R. L., Wang, T. J., Rittenhouse, H. G., Ratliff, T. L. et al: Evaluation of percentage of free serum prostate-specific antigen to improve specificity of prostate cancer screening. *JAMA*, **274**: 1214, 1995
- Catalona, W. J., Partin, A. W., Slawin, K. M., Brawer, M. K., Flanigan, R. C., Patel, A. et al: Use of the percentage of free prostate-specific antigen to enhance differentiation of prostate

- cancer from benign prostatic disease: a prospective multicenter clinical trial. *JAMA*, **279**: 1542, 1998
26. Chen, Y. T., Luderer, A. A., Thiel, R. P., Carlson, G., Cuny, C. L. and Soriano, T. F.: Using proportions of free to total prostate-specific antigen, age, and total prostate-specific antigen to predict the probability of prostate cancer. *Urology*, **47**: 518, 1996
 27. Carter, H. B., Pearson, J. D., Metter, E. J., Brant, L. J., Chan, D. W., Andres, R. et al: Longitudinal evaluation of prostate-specific antigen levels in men with and without prostate disease. *JAMA*, **267**: 2215, 1992
 28. Smith, D. S. and Catalona, W. J.: Rate of change of serum prostate specific antigen levels as a method for prostate cancer detection. *J Urol*, **152**: 1163, 1994
 29. Nadler, R. B., Humphrey, P. A., Smith, D. S., Catalona, W. J. and Ratliff, T. L.: Effect of inflammation and benign prostatic hyperplasia on elevated serum prostate specific antigen levels. *J Urol*, **154**: 407, 1995
 30. Seaman, E., Whang, M., Olsson, C. A., Katz, A., Cooner, W. H. and Benson, M. C.: PSA density (PSAD). Role in patient evaluation and management. *Urol Clin North Am*, **20**: 653, 1993
 31. Bare, R., Hart, L. and McCullough, D. L.: Correlation of prostate-specific antigen and prostate-specific antigen density with outcome of prostate biopsy. *Urology*, **43**: 191, 1994
 32. Brawer, M. K., Aramburu, E. A. G., Chen, G. L., Preston, S. D. and Ellis, W. J.: The inability of prostate specific antigen index to enhance the predictive value of prostate specific antigen in the diagnosis of prostatic carcinoma. *J Urol*, **150**: 369, 1993
 33. Catalona, W. J., Southwick, P. C., Slawin, K. M., Partin, A. W., Brawer, M. K., Flanigan, R. C. et al: Comparison of percent free PSA, PSA density, and age-specific PSA cutoffs for prostate cancer detection and staging. *Urology*, **56**: 255, 2000
 34. Catalona, W. J., Ramos, C. G., Carvalhal, G. F. and Yan, Y.: Lowering PSA cutoffs to enhance detection of curable prostate cancer. *Urology*, **55**: 791, 2000
 35. Catalona, W. J., Smith, D. S. and Ornstein, D. K.: Prostate cancer detection in men with serum PSA concentrations of 2.6 to 4.0 ng/ml and benign prostate examination. Enhancement of specificity with free PSA measurements. *JAMA*, **277**: 1452, 1997
 36. Smith, D. S., Carvalhal, G. F., Mager, D. E., Bullock, A. D. and Catalona, W. J.: Use of lower prostate specific antigen cutoffs for prostate cancer screening in black and white men. *J Urol*, **160**: 134, 1998
 37. Smith, D. S., Catalona, W. J. and Herschman, J. D.: Longitudinal screening for prostate cancer with prostate-specific antigen. *JAMA*, **276**: 1309, 1996
 38. Gann, P. H., Hennekens, C. H. and Stampfer, M. J.: A prospective evaluation of plasma prostate-specific antigen for detection of prostate cancer. *JAMA*, **273**: 289, 1995
 39. Carter, H. B.: A PSA threshold of 4.0 ng/mL for early detection of prostate cancer: the only rational approach for men 50 years old and older. *Urology*, **55**: 796, 2000
 40. Carter, H. B., Epstein, J. I. and Partin, A. W.: Influence of age and prostate-specific antigen on the chance of curable prostate cancer among men with nonpalpable disease. *Urology*, **53**: 126, 1999
 41. Carey, J. M. and Korman, H. J.: Transrectal ultrasound guided biopsy of the prostate. Do enemas decrease clinically significant complications? *J Urol*, **166**: 82, 2001
 42. Rodríguez, L. V. and Terris, M. K.: Risks and complications of transrectal ultrasound guided prostate needle biopsy: a prospective study and review of the literature. *J Urol*, **160**: 2115, 1998
 43. Kapoor, D. A., Klimberg, I. W., Malek, G. H., Wegenke, J. D., Cox, C. E., Patterson, A. L. et al: Single-dose oral ciprofloxacin versus placebo for prophylaxis during transrectal prostate biopsy. *Urology*, **52**: 552, 1998
 44. Shandera, K. C., Thibault, G. P. and Deshon, G. E., Jr.: Variability in patient preparation for prostate biopsy among American urologists. *Urology*, **52**: 644, 1998
 45. Clements, R., Aideyan, O. U., Griffiths, G. J. and Peeling, W. B.: Side effects and patient acceptability of transrectal biopsy of the prostate. *Clin Radiol*, **47**: 125, 1993
 46. Collins, G. N., Lloyd, S. N., Hehir, M. and McKelvie, G. B.: Multiple transrectal ultrasound-guided prostatic biopsies—true morbidity and patient acceptance. *Br J Urol*, **71**: 460, 1993
 47. Nash, P. A., Bruce, J. E., Indudhara, R. and Shinohara, K.: Transrectal ultrasound guided prostatic nerve blockade eases systematic needle biopsy of the prostate. *J Urol*, **155**: 607, 1996
 48. Issa, M. M., Bux, S., Chun, T., Petros, J. A., Labadia, A. J., Anastasia, K. et al: A randomized prospective trial of intrarectal lidocaine for pain control during transrectal prostate biopsy: the Emory University experience. *J Urol*, **164**: 397, 2000
 49. Alavi, A. S., Soloway, M. S., Vaidya, A., Lynne, C. M. and Gheiler, E. L.: Local anesthesia for ultrasound guided prostate biopsy: a prospective randomized trial comparing 2 methods. *J Urol*, **166**: 1343, 2001
 50. Eskew, L. A., Bare, R. L. and McCullough, D. L.: Systematic 5 region prostate biopsy is superior to sextant method for diagnosing carcinoma of the prostate. *J Urol*, **157**: 199, 1997
 51. Stewart, C. S., Leibovich, B. C., Weaver, A. L. and Lieber, M. M.: Prostate cancer diagnosis using a saturation needle biopsy technique after previous negative sextant biopsies. *J Urol*, **166**: 86, 2001
 52. Hodge, K. K., McNeal, J. E., Terris, M. K. and Stamey, T. A.: Random systematic versus directed ultrasound guided transrectal core biopsies of the prostate. *J Urol*, **142**: 71, 1989
 53. Levine, M. A., Ittman, M., Melamed, J. and Lepor, H.: Two consecutive sets of transrectal ultrasound guided sextant biopsies of the prostate for the detection of prostate cancer. *J Urol*, **159**: 471, 1998
 54. Stamey, T. A.: Making the most out of six systematic sextant biopsies. *Urology*, **45**: 2, 1995
 55. Babaian, R. J., Toi, A., Kamoi, K., Troncoso, P., Sweet, J., Evans, R. et al: A comparative analysis of sextant and an extended 11-core multisite directed biopsy strategy. *J Urol*, **163**: 152, 2000
 56. Presti, J. C., Jr., Chang, J. J., Bhargava, V. and Shinohara, K.: The optimal systematic prostate biopsy scheme should include 8 rather than 6 biopsies: results of a prospective clinical trial. *J Urol*, **163**: 163, 2000
 57. Naughton, C. K., Miller, D. C., Mager, D. E., Ornstein, D. K. and Catalona, W. J.: A prospective randomized trial comparing 6 versus 12 prostate biopsy cores: impact on cancer detection. *J Urol*, **164**: 388, 2000
 58. Eskew, L. A., Woodruff, R. D., Bare, R. L. and McCullough, D. L.: Prostate cancer diagnosed by the 5 region biopsy method is significant disease. *J Urol*, **160**: 794, 1998
 59. Chan, T. Y., Chan, D. Y., Lecksell, K., Stutzman, R. E. and Epstein, J. I.: Does increased needle biopsy sampling of the prostate detect a higher number of potentially insignificant tumors? *J Urol*, **166**: 2181, 2001
 60. Applewhite, J. C., Matlaga, B. R. and McCullough, D. L.: Results of the 5 region prostate biopsy method: the repeat biopsy population. *J Urol*, **168**: 500, 2002
 61. Uzzo, R. G., Wei, J. T., Waldbaum, R. S., Perlmutter, A. P., Byrne, J. C. and Vaughan, E. D., Jr.: The influence of prostate size on cancer detection. *Urology*, **46**: 831, 1995
 62. Karakiewicz, P. I., Bazinet, M., Aprikian, A. G., Trudel, C., Aronson, S., Nachabe, M. et al: Outcome of sextant biopsy according to gland volume. *Urology*, **49**: 55, 1997
 63. Vashi, A. R., Wojno, K. J., Gillespie, B. and Oesterling, J. E.: A model for the number of cores per prostate biopsy based on patient age and prostate gland volume. *J Urol*, **159**: 920, 1998
 64. Stricker, H. J., Ruddock, L. J., Wan, J. and Belville, W. D.: Detection of non-palpable prostate cancer. A mathematical and laboratory model. *Br J Urol*, **71**: 43, 1993
 65. Applewhite, J. C., Matlaga, B. R. and McCullough, D. L.: Results of the 5 region prostate biopsy method: the repeat biopsy population. *J Urol*, **168**: 500, 2002
 66. Epstein, J. I., Walsh, P. C., Akingba, G. and Carter, H. B.: The significance of prior benign needle biopsies in men subsequently diagnosed with prostate cancer. *J Urol*, **162**: 1649, 1999
 67. Ellis, W. J. and Brawer, M. K.: Repeat prostate needle biopsy: who needs it? *J Urol*, **153**: 1496, 1995
 68. Keetch, D. W., Catalona, W. J. and Smith, D. S.: Serial prostatic biopsies in men with persistently elevated serum prostate specific antigen values. *J Urol*, **151**: 1571, 1994
 69. Fleshner, N. E., O'Sullivan, M. and Fair, W. R.: Prevalence and predictors of a positive repeat transrectal ultrasound guided needle biopsy of the prostate. *J Urol*, **158**: 505, 1997
 70. Borboroglu, P. G., Comer, S. W., Riffenburgh, R. H. and Amling, J. P.: Repeat prostate needle biopsy: who needs it? *J Urol*, **153**: 1496, 1995

- C. L.: Extensive repeat transrectal ultrasound guided prostate biopsy in patients with previous benign sextant biopsies. *J Urol*, **163**: 158, 2000
71. Perrotti, M., Han, K.-R., Epstein, R. E., Kennedy, E. C., Rabbani, F., Badani, K. et al: Prospective evaluation of endorectal magnetic resonance imaging to detect tumor foci in men with prior negative prostatic biopsy: a pilot study. *J Urol*, **162**: 1314, 1999
 72. Bazinet, M., Karakiewicz, P. I., Aprikian, A. G., Trudel, C., Aronson, S., Nachabé, M. et al: Value of systematic transition zone biopsies in the early detection of prostate cancer. *J Urol*, **155**: 605, 1996
 73. Terris, M. K., Pham, T. Q., Issa, M. M. and Kabalin, J. N.: Routine transition zone and seminal vesicle biopsies in all patients undergoing transrectal ultrasound guided prostate biopsies are not indicated. *J Urol*, **157**: 204, 1997
 74. Fleshner, N. E. and Fair, W. R.: Indications for transition zone biopsy in the detection of prostatic carcinoma. *J Urol*, **157**: 556, 1997
 75. Epstein, J. I., Walsh, P. C., Sauvegeot, J. and Carter, H. B.: Use of repeat sextant and transition zone biopsies for assessing extent of prostate cancer. *J Urol*, **158**: 1886, 1997
 76. Keetch, D. W. and Catalona, W. J.: Prostatic transition zone biopsies in men with previous negative biopsies and persistently elevated serum prostatic specific antigen values. *J Urol*, **154**: 1795, 1995
 77. Lui, P. D., Terris, M. K., McNeal, J. E. and Stamey, T. A.: Indications for ultrasound guided transition zone biopsies in the detection of prostate cancer. *J Urol*, **153**: 1000, 1995
 78. Chen, M. E., Troncoso, P., Tang, K., Babaian, R. J. and Johnston, D.: Comparison of prostate biopsy schemes by computer simulation. *Urology*, **53**: 951, 1999
 79. Seaman, E. K., Sawczuk, I. S., Fatal, M., Olsson, C. A. and Shabsigh, R.: Transperineal prostate needle biopsy guided by transurethral ultrasound in patients without a rectum. *Urology*, **47**: 353, 1996
 80. Shinghal, R. and Terris, M. K.: Limitations of transperineal ultrasound-guided prostate biopsies. *Urology*, **54**: 706, 1999
 81. Norberg, M., Holmberg, L., Haggman, M. and Magnusson, A.: Determinants of complications after multiple transrectal core biopsies of the prostate. *Eur Radiol*, **6**: 457, 1996
 82. Brullet, E., Guevara, M. C., Campo, R., Falco, J., Puig, J., Prera, A. et al: Massive rectal bleeding following transrectal ultrasound-guided prostate biopsy. *Endoscopy*, **32**: 792, 2000